Complex Numbers (Algebra)

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Contents

Part I

Theory

Chapter 1

Complex Numbers

Matrix 1: A line $\overline{a}z + a\overline{z} + c + \overline{c} = 0$ is taken in the argand plane. The equation of line after applying the transformation given, is to be found. Column I contains the transformation while column II contains the resulting equations.

Column I	Column II
(P)The line is translated a distance $Re(a)$ along the positive x-axis , $Im(a)$ along positive y-axis , and then reflected in the real axis	(A)$\overline{az} + az = c + \overline{c} + 2Re(ia^2)$
(Q)The line is translated a distance $Re(a)$ along the positive x-axis , $Im(a)$ along positive y-axis , and then reflected in imaginary axis	(B)$\overline{az} + az + c + \overline{c} = 0$
(P)The line is translated a distance $Im(a)$ along the positive x-axis , $Re(a)$ along negative y-axis , and then reflected in the real axis	(C)$\overline{az} + az + c + \overline{c} - 2\left\|a\right\|^2 = 0$
(P)The line is translated a distance $Im(a)$ along the positive x-axis , $Re(a)$ along positive y-axis , and then reflected in imaginary axis	(D)$\overline{az} + az + 2\left\|a\right\|^2 = c + \overline{c}$
	(E) $\overline{az} + az + 2Im(a^2) = c + \overline{c}$

Part II

Problem Set

Chapter 2

JEE Mains Problems

Q1: Let x and y be real numbers such that $50\left(\frac{2x}{1+3i}-\frac{y}{1-2i}\right)=31+17i$, $i=\sqrt{-1}$.Then the value of $10(x-3y)$ is :

[JEE Mains 2nd Apr 2026 Shift 1 Q2]

(A) 20

(B) 31

(C) 35

(D) 75

{ Solution: https://www.youtube.com/watch?v=7bvmAsRoeiU }

Chapter 3

JEE Advanced Problems

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $z1{=}1{+}2i$ and $z2{=}3i$ be two complex numbers, where $i{=}\sqrt{-1}$. Let

$S{=}\{(x{,}y){\in}\mathbb{R}{\times}\mathbb{R}|x + iy - z_1| = 2|x + iy - z_2|\}$.

Then which of the following statements is (are) TRUE?

[JEE Advanced 2025 Paper 1 Q7]

(A) S is a circle with centre $(-\frac{1}{3}, \frac{10}{3})$

(B) S is a circle with centre $(\frac{1}{3}, \frac{8}{3})$

(C) S is a circle with radius $\frac{\sqrt{2}}{3}$

(D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

{ Solution : https://www.youtube.com/watch?v=cYJl2HBn030 }